Designing a Universal Information Language

Oscar Bender-Stone

March 26, 2026

DRAFT

Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

DRAFT

Background

- Information Management: studies how to **store** and **organize** data

DRAFT

[1]Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2]Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3]Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data

DRAFT

[1]Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2]Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3]Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data
  - Throughout *all* history: so much data!
  - Organize through **information**

DRAFT

[1]Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2]Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3]Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data
  - Throughout *all* history: so much data!
  - Organize through **information**
- What *is* information?

DRAFT

[1] Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2] Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3] Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data
  - Throughout *all* history: so much data!
  - Organize through **information**
- What *is* information?
  - Depends on who you ask!

DRAFT

[1] Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2] Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3] Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data
  - Throughout *all* history: so much data!
  - Organize through **information**
- What *is* information?
  - Depends on who you ask!
  - **Ackoff:** Data-to-Wisdom Hierarchy[1]

DRAFT

[1] Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2] Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3] Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data
  - Throughout *all* history: so much data!
  - Organize through **information**
- What *is* information?
  - Depends on who you ask!
  - **Ackoff:** Data-to-Wisdom Hierarchy[1]
    - Data ⇒ Information ⇒ Wisdom

DRAFT

[1] Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2] Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3] Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data
  - Throughout *all* history: so much data!
  - Organize through **information**
- What *is* information?
  - Depends on who you ask!
  - **Ackoff:** Data-to-Wisdom Hierarchy[1]
    - Data ⇒ Information ⇒ Wisdom
  - **Buckland:** Based on usage[2]

DRAFT

[1] Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2] Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3] Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data
  - Throughout *all* history: so much data!
  - Organize through **information**
- What *is* information?
  - Depends on who you ask!
  - **Ackoff:** Data-to-Wisdom Hierarchy[1]
    - Data ⇒ Information ⇒ Wisdom
  - **Buckland:** Based on usage[2]
    - Can be a *process*, or *knowledge*
    - Most useful as *thing*
  - **Floridi:** Information is **truthful**[3]

DRAFT

[1] Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2] Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3] Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Background

- Information Management: studies how to **store** and **organize** data
  - Throughout *all* history: so much data!
  - Organize through **information**
- What *is* information?
  - Depends on who you ask!
  - **Ackoff:** Data-to-Wisdom Hierarchy[1]
    - Data ⇒ Information ⇒ Wisdom
  - **Buckland:** Based on usage[2]
    - Can be a *process*, or *knowledge*
    - Most useful as *thing*
  - **Floridi:** Information is **truthful**[3]

DRAFT

[1] Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
[2] Buckland, M. K. (1991). *Information as thing*. JASIS, 42(5), 351–360.
[3] Floridi, L. (2010). *Information: A Very Short Introduction*. Oxford University Press.

Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]

DRAFT

[1]Smith & Linden (2017). Two Decades of Recommender Systems at Amazon.com. IEEE Internet Comput. 21(3).
[2]Mesgari, M., et al. (2015). "The sum of all human knowledge": A systematic review of Wikipedia. JASIST, 66(2).
[3]Reisman, M. (2017). EHRs: The challenge of making electronic data usable. P & T, 42(9), 572–575.

Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data

DRAFT

[1]Smith & Linden (2017). Two Decades of Recommender Systems at Amazon.com. IEEE Internet Comput. 21(3).
[2]Mesgari, M., et al. (2015). "The sum of all human knowledge": A systematic review of Wikipedia. JASIST, 66(2).
[3]Reisman, M. (2017). EHRs: The challenge of making electronic data usable. P & T, 42(9), 572–575.

Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data
  - Example: Wikipedia[2]

DRAFT

[1]Smith & Linden (2017). Two Decades of Recommender Systems at Amazon.com. IEEE Internet Comput. 21(3).
[2]Mesgari, M., et al. (2015). "The sum of all human knowledge": A systematic review of Wikipedia. JASIST, 66(2).
[3]Reisman, M. (2017). EHRs: The challenge of making electronic data usable. P & T, 42(9), 572–575.

Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data
  - Example: Wikipedia[2]
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
  - **Interoperable:** a portable data format
  - Example: US health records[3]

DRAFT

[1]Smith & Linden (2017). Two Decades of Recommender Systems at Amazon.com. IEEE Internet Comput. 21(3).
[2]Mesgari, M., et al. (2015). "The sum of all human knowledge": A systematic review of Wikipedia. JASIST, 66(2).
[3]Reisman, M. (2017). EHRs: The challenge of making electronic data usable. P & T, 42(9), 572–575.

Existing Solutions

- **Resource Description Language (RDF)**
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - − No built-in logic engine
- **Labeled Property Graphs (LPGs)**
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - − Not standardized
- **Cyc**
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - − Hard-coded, proprietary rules

DRAFT

Proposed Solution

DRAFT

Proposed Solution

- **Welkin**: an information language

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**
  - **Universality:** be completely expressive

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**
  - **Universality:** be completely expressive
    - *Any* computer program

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**
  - **Universality:** be completely expressive
    - *Any* computer program
    - *Any* “reliable” proof

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**
  - **Universality:** be completely expressive
    - *Any* computer program
    - *Any* “reliable” proof
  - **Scalability:** work at any size

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**
  - **Universality:** be completely expressive
    - *Any* computer program
    - *Any* “reliable” proof
  - **Scalability:** work at any size
    - Be efficient

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**
  - **Universality:** be completely expressive
    - *Any* computer program
    - *Any* “reliable” proof
  - **Scalability:** work at any size
    - Be efficient
    - Support: future optimizations
    - Describe improvements *in* Welkin

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**
  - **Universality:** be completely expressive
    - *Any* computer program
    - *Any* “reliable” proof
  - **Scalability:** work at any size
    - Be efficient
    - Support: future optimizations
    - Describe improvements *in* Welkin
  - **Standardization:** be completely *unambiguous*

DRAFT

Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”
- **Goals:**
  - **Universality:** be completely expressive
    - *Any* computer program
    - *Any* “reliable” proof
  - **Scalability:** work at any size
    - Be efficient
    - Support: future optimizations
    - Describe improvements *in* Welkin
  - **Standardization:** be completely *unambiguous*
    - Have a formal specification

DRAFT

Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

DRAFT

General Requirements

- Need at least two things:

DRAFT

General Requirements

- Need at least two things:
  1. Express any computer program

DRAFT

General Requirements

- Need at least two things:
  1. Express any computer program
  2. Enable user-defined *meaning*

DRAFT

Base Operations

DRAFT

Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

DRAFT

Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

DRAFT

Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

DRAFT